

## Article

# Open-Source High-Throughput Phenotyping for Blueberry Yield and Maturity Prediction Across Environments: Neural Network Model and Labeled Dataset for Breeders

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**Abstract:** Time to maturity and yield are important traits for highbush blueberry (*Vaccinium corymbosum*) breeding. Proper determination of the time to maturity of blueberry varieties and breeding lines informs the harvest window, ensuring that the fruits are harvested at optimum maturity and quality. On the other hand, high-yielding crops bring in high profits per acre of planting. Harvesting and quantifying the yield for each blueberry breeding accession are labor-intensive and impractical. Instead, visual ratings as an estimation of yield are often used as a faster way to quantify the yield, which is categorical and subjective. In this study, we developed and shared a high-throughput phenotyping method using neural networks to predict blueberry time to maturity and to provide a proxy for yield, overcoming the labor constraints of obtaining high-frequency data. We aim to facilitate further research in computer vision and precision agriculture by publishing the labeled image dataset and the trained model. In this research, true-color images of blueberry bushes were collected, annotated, and used to train a deep neural network object detection model [You Only Look Once (YOLOv11)] to detect mature and immature berries. Different versions of YOLOv11 were used, including nano, small, and medium, which had similar performance, while the medium version had slightly higher metrics. The YOLOv11m model shows strong performance for the mature berry class, with a precision of 0.90 and an F1 score of 0.90. The precision and recall for detecting immature berries were 0.81 and 0.79. The model was tested on 10 blueberry bushes by hand harvesting and weighing blueberries. The results showed that the model detects approximately 25% of the berries on the bushes, and the correlation coefficients between model-detected and hand-harvested traits were 0.66, 0.86, and 0.72 for mature fruit count, immature fruit count, and mature ratio, respectively. The model applied to 91 blueberry advance selections and categorized them into groups with diverse levels of maturity and productivity using principal component analysis (PCA). These results inform the harvest window and yield of these breeding lines with precision and objectivity through berry classification and quantification. This model will be helpful for blueberry breeders, enabling more efficient selection, and for growers, helping them accurately estimate optimal harvest windows. This open-source tool can potentially enhance research capabilities and agricultural productivity.



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**Keywords:** computer vision; YOLOv11; fruit detection; harvest window

## 1. Introduction

Blueberry (*V. corymbosum* sect. *Cyanococcus*) is a high-value fruit crop, contributing approximately USD 4.7 billion to the U.S. economy in 2020 [1]. The increasing consumer appreciation for blueberries—due to their flavor, versatility, and health benefits—has driven a 97% rise in per capita blueberry consumption over the past decade [2]. This growing demand continues to boost blueberry production across the country and elsewhere. The primary cultivated blueberry types in the U.S. include northern highbush (NHB,  $2n = 4x = 48$ ), southern highbush (SHB,  $2n = 4x = 48$ ), lowbush (LB,  $2n = 4x = 48$ ), and rabbiteye (RE,  $2n = 6x = 72$ ). The NHB and LB, both native to North America, require high chilling hours and thrive in the cooler climates of the northwest and northeast regions. The SHB, on the other hand, was developed by introgressing the low chill requirement from several diploid and hexaploid species, including *V. darrowii*, *V. fuscatum*, *V. elliotii*, *V. pallidum*, *V. virgatum* (formerly *V. ashei*), into the NHB varieties [3,4]. These cultivated blueberries exhibit significant diversity in yield, phenological development, adaptability to soil and weather, fruit quality, and plant morphology.

One of the primary goals of the blueberry breeding programs at the University of Georgia and North Carolina State University is to develop SHB varieties that are both high-yielding and have staggering harvest windows. Early maturing varieties provide growers with a valuable market advantage by allowing them to tap into premium pricing. However, the high volume of imported blueberries during the U.S. southeastern harvest season incurs market price competition. Between April and May 2024, USD 200 M of fresh blueberries was imported into the U.S. [5]. Developing SHB blueberries with staggering harvest windows will mitigate the risk of price crashes due to the high volume of production conflicting with the imported blueberries. Therefore, extending the production season by developing varieties with staggering maturity times is critical for sustaining the competitiveness of the U.S. blueberry industry. This is particularly beneficial for growers in Georgia, North Carolina, and Florida by offering greater resilience to rising temperatures and unpredictable weather patterns [6] (Ashrafi Pers. Comm.). However, as with many breeding programs, blueberry phenotyping faces challenges due to labor and time constraints. Efficient and high-throughput phenotyping methods are critically needed to assist the proper selection of breeding progenies to advance them to the next stage. Various high-throughput phenotyping techniques have been employed to enhance data quality and consistency in blueberry breeding to characterize multiple traits. For example, the canopy architecture traits of blueberry bushes were analyzed using Unmanned Aerial Systems (UASs) and point cloud construction [7]. The yield estimation of lowbush blueberries was achieved through the blue pixel index, which was extracted from RGB images captured by cameras mounted on a motorized vehicle [8]. Additionally, an instant segmentation algorithm based on Mask R-CNN [a type of Convolutional Neural Network (CNN)] was developed to accurately detect blueberry clusters from images taken with mobile devices. Lobos et al. (2019) [9] explored the use of leaf-level reflectance to assess physiological traits in five *Vaccinium* cultivars under different stress conditions, and partial least square modeling provided good trait estimates and demonstrated the potential of reflectance-based tools for assessing physiological traits under environmental stresses in blueberry breeding programs. Furthermore, hyperspectral reflectance imaging was utilized for the non-destructive detection of internal bruising in harvested berries [10]. Collectively, these high-throughput systems facilitate the quantification of phenotypic traits in blueberries, significantly accelerating the process of breeding line selection. However, there are limited or no reports on high-throughput yield and fruit maturing prediction tools.

Blueberry time to maturity and yield are important traits for both growers and breeders. Developing high-yielding cultivars with targeted maturing time suitable to the local growing environment is a priority for breeding programs. Growers need to harvest within the optimal time window to ensure the fruit's highest quality and desirable maturity. Harvesting too early can lead to excessive removal of green fruits, whereas harvesting too late results in overripened fruits that compromise the overall fruit quality [11]. Cultivars with

concentrated ripening time are desirable for growers who adopted mechanical harvesting, as they can enhance the profitability of the harvesting process. However, estimating both yield and time to maturity in blueberries presents challenges, as berries on the same bush do not ripen simultaneously during the harvest season. Blueberry time to maturity can be approximated by the ratio of purple to green fruits, with a high ratio indicating a greater level of maturity. Previously, true-color images, also referred to as RGB (red, green, and blue) images, were used to classify blueberry fruit stages. Among the four tested classifiers, the color component analysis-based detection was identified as the most effective method for distinguishing blueberry developmental stages [12]. Recently, advancements in artificial intelligence have enhanced object detection capabilities [13]. Deep neural networks have been trained to estimate berry ripeness and yield [14–19]. Previous research has primarily focused on developing novel model architectures [17,19,20] or exploring their potential applications in production systems [15,16]. Although research has proved that deep neural networks can do an excellent job counting objects [21], the model for counting blueberries with the goal of yield prediction and determining time to maturity is unavailable for small breeding programs.

Advancements in computing resources and the availability of foundational models have greatly empowered practitioners in object counting and detection tasks [22]. High-performance hardware, such as GPUs and cloud-based platforms, enable efficient training and deployment of complex models. Additionally, pre-trained models, such as those based on YOLO, Mask R-CNN, and other state-of-the-art architectures, provide a strong starting point for customization. These foundational models allow practitioners to leverage transfer learning [23], significantly reducing the time and effort required to train their models for specific applications, in our case, counting blueberries. With the increasing availability of resources for object detection and counting, human labeling has become a significant component of projects like ours [21,24]. To streamline this process and reduce redundancy, we aim to publish our labeled dataset, making it accessible to researchers and practitioners. By sharing these data, we hope to facilitate collaboration, enhance model training efficiency, and enable others to build on our work without the need for extensive manual labeling. This will accelerate research and contribute to the broader understanding of blueberry phenotyping and yield estimation.

In this study, we adapted the YOLOv11 model architecture for blueberry phenotyping, focusing on predicting yield and estimating time to maturity within two breeding programs. We developed a high-throughput phenotyping pipeline to support these objectives. Our goals are threefold: (1) to create and publish a labeled image dataset for integration into computer vision and machine learning projects; (2) to develop a robust pipeline for yield prediction and time-to-maturity estimation; and (3) to make the model publicly available for breeders, particularly those with small-sized budgets. This model will enable blueberry breeders to optimize the selection of high-yielding, early maturing varieties, and growers can use it to estimate optimal harvest windows, improving decision making and productivity.

## 2. Materials and Methods

**Study Locations and Description of the Blueberry Bushes.** Images were collected initially at the UGA Alapaha research farm, Alapaha, GA (31.3852° N, 83.2229° W, Figure 1). The blueberry breeding population at this location consists of 91 SHB breeding lines from 27 independent crossing families, selected based on the adaptability of the local growing environment and fruit quality. The advanced selections were clonally propagated by softwood cuttings and transplanted in 2018 and 2019. To generalize the model, another set of images was collected at Horticultural Crop Research Station Ideal Field located in Castle Hayne, NC (34.3648° N, 77.8387° W). A mechanical harvest trial of advanced SHB breeding selections was planted in 10–11 plant plots 1 m apart with 3 m row centers in October–November of 2019 and 2020.



**Figure 1.** Map of the locations of the blueberry orchards where images were collected.

**Image Collection, Curation, and Annotation.** In Alapaha, true-color images of the population were collected on May 12 and 23 of 2023 using a handheld digital camera (Panasonic, Lumix DMC-LX10) with a resolution of  $5472 \times 3648$ . In Castle Hayne, true-color images were collected on the yield trial on 19 May 2024, using a digital camera (Canon EOS Rebel T7) with a resolution of  $6000 \times 4000$ . The operator stood approximately 1 m away from the target bush and held the camera facing the bush with the field of vision centered on the middle of the canopy of the bush. All images were taken from one side of the bushes. The weather conditions during image collection varied between clear and overcast in both locations. Approximately 80 raw images from both locations were split  $5 \times 3$  into 15 cropped images for preparing the image dataset for training. Some pictures were cropped, in a way that the dimensions after the cut were set equal to or smaller than  $1200 \times 1200$  pixels (Figure 2). Object detection models such as YOLO and Mask R-CNN do not work well out of the box to detect small objects in high-resolution images. One common way to address this problem is to slice the image into smaller tiles before running inference [25]. The cropped images (~466) were then exported to Roboflow (<https://roboflow.com/>, accessed on 8 December 2024) for annotation with two classes (Figure 3): mature fruit (ripe) and immature fruit (green and pink). In Roboflow, Auto-Orient and Resize ( $640 \times 640$  pixels) were applied to the images in preprocessing. The following augmentations were then applied in Roboflow: flip horizontally, crop 0–20%, rotation  $-15^\circ$  to  $+15^\circ$ , saturation  $-25\%$  to  $+25\%$ , and blur up to 2.5 pixels. A total of 2686 images were exported for model training.

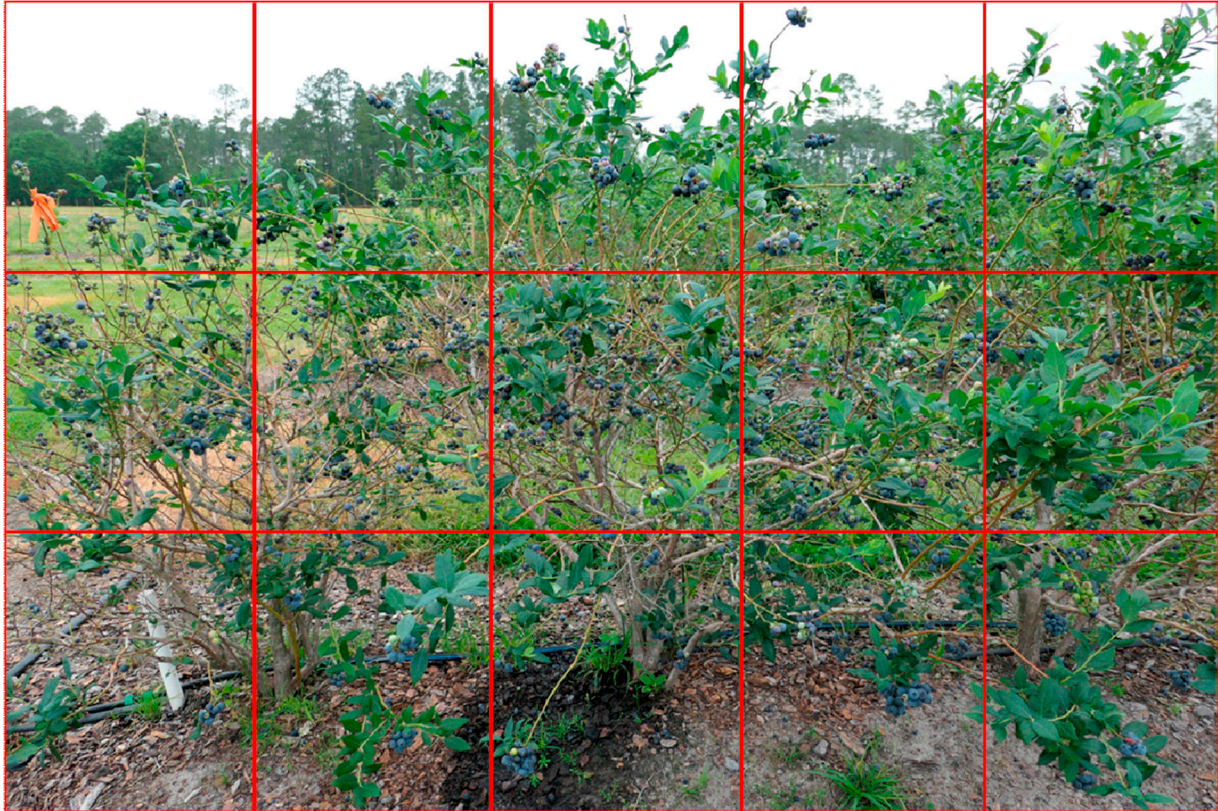
**Model Training.** Following augmentation, the dataset containing 2686 images was split into training (80%), validation (14%), and test (6%) sets. The dataset was exported and used to train three versions of YOLOv11 models: nano, small, and medium (using Ultralytics 8.3.13 [26] in PyTorch 2.4.0 [27]). The training was conducted on a workstation computer equipped with an NVIDIA GeForce RTX 4070 Ti GPU. The YOLOv11 trainer ran for 30 epochs with a batch size 8 and an image size of 680 (Figure S1). A confidence threshold of 0.4 was chosen after examining the relationship between precision and recall as the confidence threshold changes (Figure S2). To evaluate the performance of three models (nano, small, and medium) on the validation and test datasets, metrics including precision, recall, and F1 score were calculated using Equations (1)–(3) [28].

$$\text{Precision} = \frac{\text{True positive}}{(\text{True positive} + \text{False positive})} \quad (1)$$

$$\text{Recall} = \frac{\text{True positive}}{(\text{True positive} + \text{False negative})} \quad (2)$$

$$\text{F1 score} = 2 \times \frac{(\text{Precision} \times \text{Recall})}{(\text{Precision} + \text{Recall})} \quad (3)$$

where true positive represents predictions that match the true label, false positive represents the incorrect predictions upon a positive detection, and false negative means that the model failed to predict a true label. In our case, lower precision and recall in detecting green berries indicate that the model struggled to identify green berries and would likely benefit from retraining with additional green berries. Three models have similar performance with the YOLO11 medium and had slightly higher metrics; therefore, they were used for the following testing and application.



**Figure 2.** The picture is cropped into smaller sections for labeling, model training, and prediction.

**Model Testing Against Harvested Yield.** To test the model, images of another ten different SHB breeding lines were taken on 11 May 2023, before harvest. Both mature and immature berries were hand-harvested right after imaging and brought to the lab for weighing. Berries were sorted manually into mature and immature groups. The total weights of mature and immature berries from each bush were taken separately. To estimate the fruit number, the weights of twenty-five mature fruit and immature fruit were taken randomly three times. Average weight of the three measurements of 25 berries was used to estimate the harvested berry numbers following Equations (4) and (5) for mature and immature berries, respectively.

$$\frac{\text{weight of total mature berries}}{\text{average weight of 25 mature berries}} \times 25 \quad (4)$$

$$\frac{\text{weight of total immature berries}}{\text{average weight of 25 immature berries}} \times 25 \quad (5)$$

Fruit numbers of the ten bushes were predicted using the object detection model (YOLO11 medium) with images cropped around  $680 \times 680$ , and the fruit numbers were summed across cropped images belonging to the same bush. Mature ratio was computed by dividing the mature fruit count by the total fruit count. Pearson correlation between the

weight-based and model-predicted berry numbers and the mature ratio was performed in R [29].



**Figure 3.** Cropped images and their annotations of two classes: mature (purple box) and immature fruits (yellow box). The image contains 184 mature and 48 immature annotations.

**Model Application.** To evaluate the efficacy of the model for phenotyping blueberry breeding accessions, 455 images of individual bushes from 91 blueberry breeding genotypes were taken on 12 May 2023, with each accession having five bushes. The object detection model was applied to detect mature and immature berries from these images. Mature fruit ratio was computed by dividing the number of mature fruits by total fruits counts. For each image, a visual rating of mature fruit ratio was given by a trained technician on a scale of 1 to 6, with 1 being 0% mature fruit, 3 being 40% mature fruit, and 6 being 100% mature fruit. This scale is used in the UGA blueberry breeding program to rate the maturity ratio. Visual ratings from two images per line were averaged. The correlation was determined between model-based and visually rated mature fruit ratios using Pearson correlation in R. Principal component analysis was conducted on fruit counts and mature ratios from both model prediction and visual ratings using package FactoMineR [30], and the associated graphs were made using package factoextra [31] in R.

**Availability of the Labeled Images and Model.** Labeled images are published in Zenodo [32], an open access repository supported by the European Organization for Nuclear Research (CERN). The model is available on GitHub [33]. To optimize the model's effectiveness, it is suggested that images be captured on overcast days or using polarized lenses to minimize glare and maintain a consistent distance from the bushes. Depending on the camera specification, images are suggested to be cropped into  $700 \times 700$  pixels for inference. A Graphical User Interface (GUI) is being developed for running the model on counting blueberries for a single image and for a batch of images. Once developed, it will be published on GitHub.

### 3. Results and Discussion

#### 3.1. Model Performance on the Validation and Test Datasets

Metrics, including precision, recall, and F1 score, indicated similar performance from three models, with YOLO11m having slightly better performance (Table 1). The model shows strong performance for the mature berry class, correctly identifying 91% of ripe

blueberries, with precision ranging from 0.85 to 0.90 and F1 score ranging from 0.88 to 0.90. The model correctly identifies 72–79% of the green berries, with precision ranging from 0.78 to 0.84 and F1 score ranging from 0.76 to 0.80. Our precision in detecting mature berries is comparable to another study, in which similar images were used to train an improved YOLOv8 model to achieve 0.86 precision in detecting ripe blueberries [34]. This demonstrated that the model performs well in identifying mature berries with high precision and minimal misclassification, demonstrating its potential for practical application in detecting ripe blueberries. While there are some challenges in distinguishing green berries from the background, the model still performs reasonably well in those categories. The model can be even more effective with further refinement, particularly in enhancing the distinction between green berry and background. Although other methods, such as color component analysis [12] and HOG-SVM classifiers [35], may report higher precision, YOLO-based models offer key advantages in speed, efficiency, and automatic feature extraction.

**Table 1.** Model performance is indicated by metrics when models predict on both validation and test datasets. Three YOLO models were used as pre-trained models and confidence of 0.4 was used.

| Validation |           |               |          |           |                 |          |
|------------|-----------|---------------|----------|-----------|-----------------|----------|
|            | Precision | Mature Recall | F1 score | Precision | Immature Recall | F1 score |
| YOLO11n    | 0.85      | 0.91          | 0.88     | 0.84      | 0.72            | 0.78     |
| YOLO11s    | 0.89      | 0.91          | 0.90     | 0.82      | 0.76            | 0.79     |
| YOLO11m    | 0.90      | 0.91          | 0.90     | 0.81      | 0.79            | 0.80     |
| Test       |           |               |          |           |                 |          |
|            | Precision | Mature Recall | F1 score | Precision | Immature Recall | F1 score |
| YOLO11n    | 0.85      | 0.91          | 0.88     | 0.82      | 0.70            | 0.76     |
| YOLO11s    | 0.88      | 0.91          | 0.89     | 0.82      | 0.77            | 0.79     |
| YOLO11m    | 0.89      | 0.91          | 0.90     | 0.78      | 0.76            | 0.77     |

### 3.2. Model Testing

The results in Table 2 highlight the model's ability to detect blueberries across 10 different breeding lines. On average, the model detected 25% mature berries, ranging from 16% to 34%, and 26% immature berries, with a range of 12% to 44%. The model's predictions correlated well with hand-harvested data, showing an  $r$  value of 0.66 for mature berries and 0.86 for immature (green) berries. Several factors may have impacted the model's performance, including images capturing only one side of the bush and some berries being obscured by leaves. Variations in detection across the breeding lines could be attributed to genotypic differences in growth stage, canopy architecture, and foliage density. Occlusion by other objects in blueberry fruit detection has been a common issue reported previously [36]. The average mature/immature ratio detected by the model (0.84) was closely aligned with the hand-harvested data (0.84). Additionally, a significant correlation ( $r = 0.72$ ) between model-detected and hand-harvested mature ratios highlights the model's potential for accurately estimating fruit maturity, particularly in production systems with limited genotypic variation.

Since only one image was taken for each bush, the berry detection by the model was limited to the area covered by the image, whereas the hand-harvested berry count included all berries produced by the bush. As a result, only about 25% of the berries were captured through image analysis. This percentage is important information and can serve as a reference and baseline for further improvements in yield prediction. Despite this limitation, the strong correlation for both mature and immature berries indicates that 2D images taken with handheld cameras can reliably estimate blueberry yield and differentiate the productivity of various blueberry breeding lines. In this study, it took about 2 h for three people to hand-harvest 30,815 berries from the ten bushes, equating to an average of 12 min for one person to harvest 1000 berries. One of the greatest challenges in blueberry

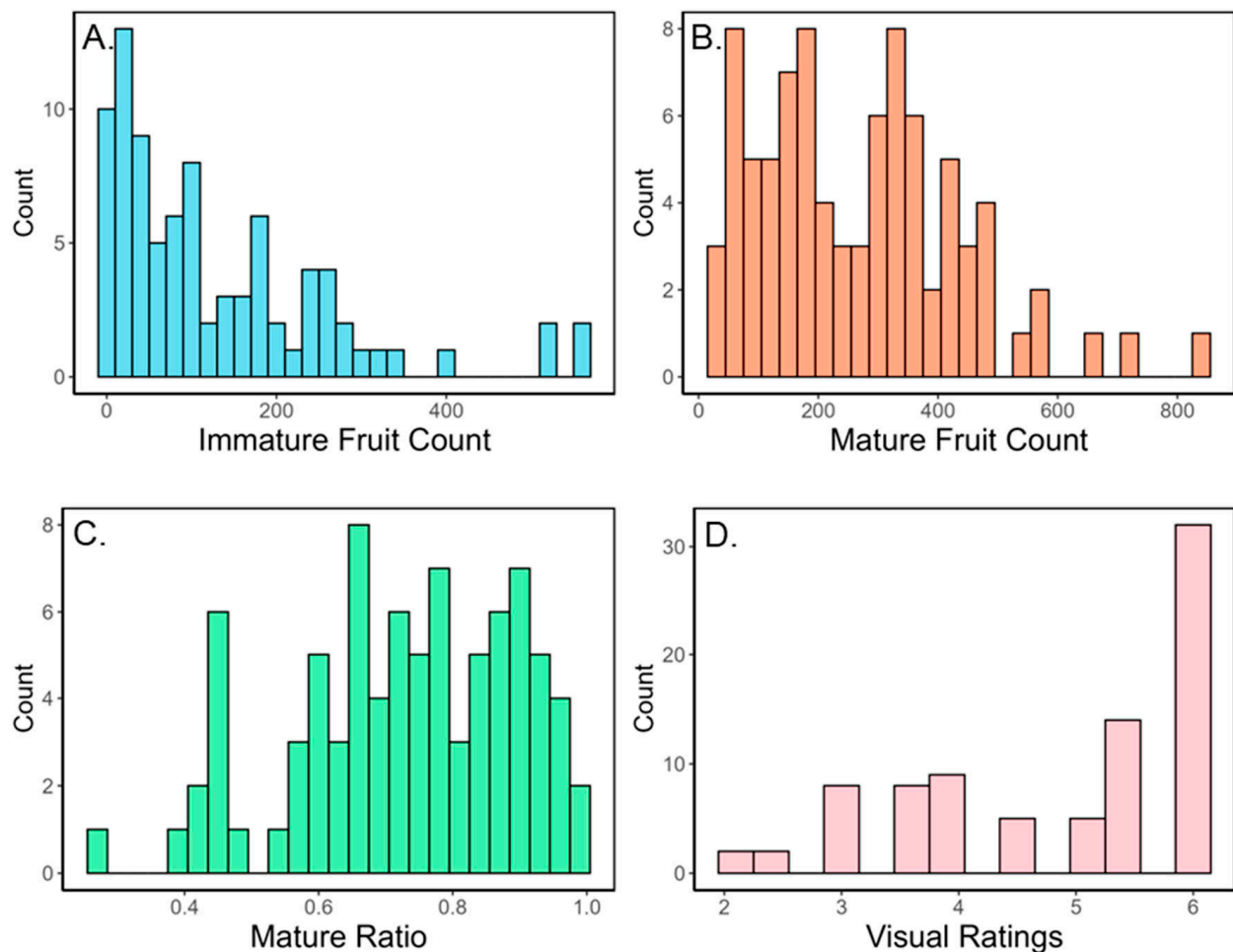
breeding programs is the limited availability of labor during the fruiting season. The high labor demand required for manual berry harvesting and yield evaluation makes collecting data for all breeding lines in the program impractical. Therefore, breeders often rely on a visual scale of 1 to 10, with 1 being the least desirable and 10 being the most desirable, to assess fruit load on the bushes [37]. The main advantage of using single-view images in blueberry phenotyping is that it significantly reduces the time and resources needed for image collection and processing while still providing reasonably accurate results. Although multi-view imaging improves the detection accuracy by 5.2% to 15.7% [36], single-view images can still achieve reliable maturity and yield estimates, especially in environments with consistent canopy architecture and minimal occlusion. This approach is particularly instrumental when rapid assessments are needed in large-scale production or breeding programs. Imaging analysis provides a high-throughput, objective alternative to determine crop load, accelerating the selection process in breeding programs.

**Table 2.** Model-predicted results on counting mature fruit and green fruit, mature ratio compared to hand-harvest results.

| Genotype | Mature Fruit Count |       |            | Green Fruit Count |       |            | Mature Ratio |       |
|----------|--------------------|-------|------------|-------------------|-------|------------|--------------|-------|
|          | Hand               | Model | % detected | Hand              | Model | % detected | Hand         | Model |
| 1        | 1755               | 535   | 30%        | 416               | 181   | 44%        | 0.81         | 0.75  |
| 2        | 3174               | 760   | 24%        | 365               | 95    | 26%        | 0.90         | 0.89  |
| 3        | 3072               | 1053  | 34%        | 1181              | 422   | 36%        | 0.72         | 0.71  |
| 4        | 2362               | 507   | 21%        | 627               | 157   | 25%        | 0.79         | 0.76  |
| 5        | 3601               | 1000  | 28%        | 527               | 94    | 18%        | 0.87         | 0.91  |
| 6        | 2221               | 634   | 29%        | 759               | 92    | 12%        | 0.75         | 0.87  |
| 7        | 1101               | 365   | 33%        | 372               | 56    | 15%        | 0.75         | 0.87  |
| 8        | 3927               | 614   | 16%        | 195               | 52    | 27%        | 0.95         | 0.92  |
| 9        | 1774               | 372   | 21%        | 299               | 77    | 26%        | 0.86         | 0.83  |
| 10       | 3015               | 489   | 16%        | 73                | 20    | 28%        | 0.98         | 0.96  |
| Ave.     | 2600               | 633   | 25%        | 481               | 125   | 26%        | 0.84         | 0.84  |
| r        | 0.66               |       |            | 0.86              |       |            | 0.72         |       |

3.3. Model Application for Mature Fruit Ratio and Harvest Window in Breeding Lines

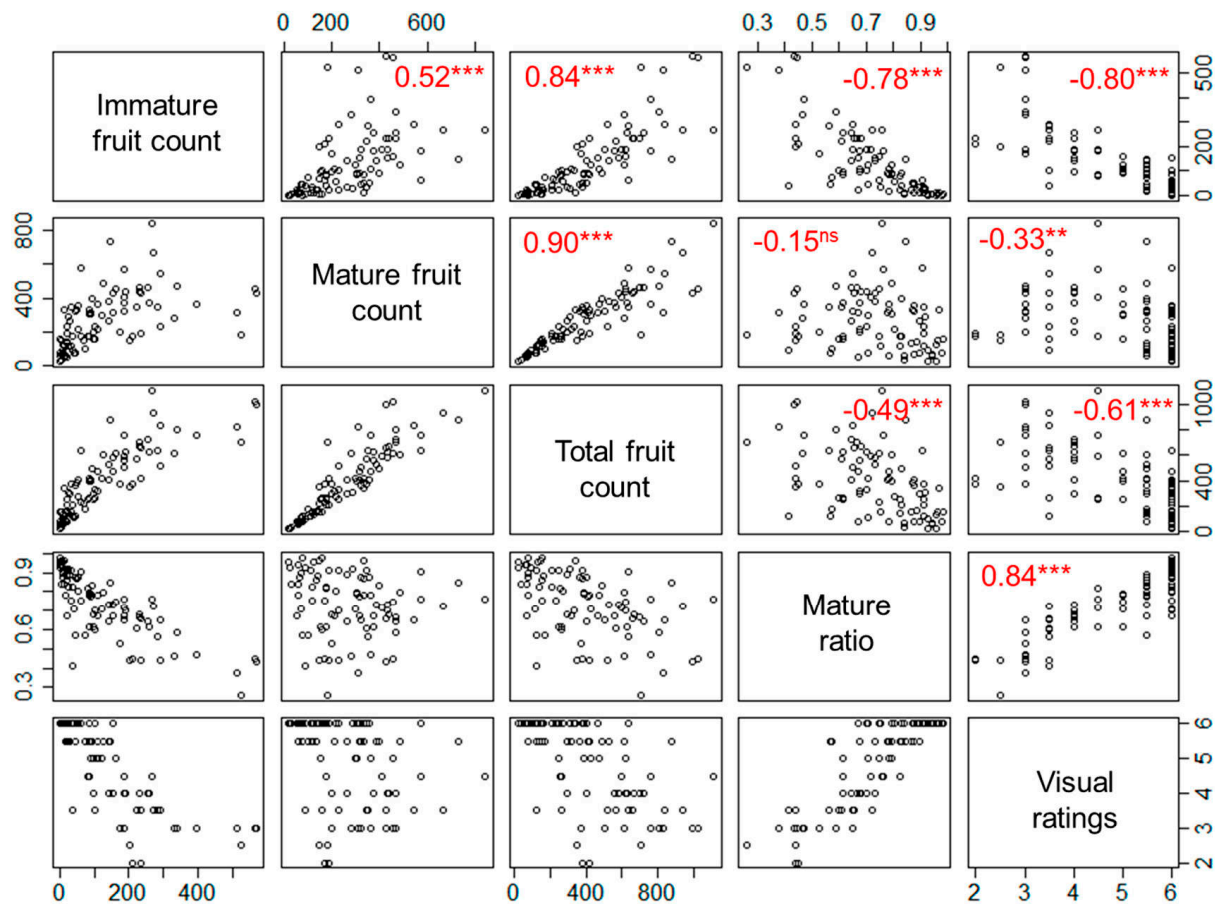
We further applied fruit detection on 91 breeding advanced selections to detect differential fruit yield and maturation. A large variation in the fruit yield across the breeding genotypes was detected, which was influenced by yield potential and the phenological development of each advanced selection (Figure 4). The histograms present insights into the distribution of fruit counts and maturity ratios detected by the model and visual ratings across different blueberry bushes (Figure 4). The distribution of immature fruit counts is right-skewed (Figure 4A), with most bushes having fewer than 100 immature fruits. However, a few outliers show higher counts. In contrast, the mature fruit count is more symmetrically distributed (Figure 4B), with most bushes producing between 100 and 400 mature fruits. The mature ratio shows that most bushes have a high proportion of mature fruits (Figure 4C), with ratios clustering between 0.6 and 0.9, indicating favorable maturity levels. The visual ratings are discrete data (Figure 4D), which may not be adequate to capture the full spectrum of variations among the breeding lines. In contrast, quantitative data derived from measurements such as fruit counts and maturity ratios offer a more comprehensive understanding of blueberry phenotyping, allowing breeders to make informed decisions based on precise metrics rather than subjective assessments.



**Figure 4.** Histogram showing distributions of (A) immature fruit count, (B) mature fruit count, (C) mature ratio predicted from the model, and (D) visual ratings of mature ratio on a 1 to 6 scale in a blueberry population consisting of 91 breeding lines in Alapaha, GA, 2023.

Significant correlations among several variables related to fruit counts, maturity ratios, and visual ratings were found (Figure 5). A strong positive and significant correlation between total fruit count and both mature fruit count ( $r = 0.90$ ,  $p < 0.05$ ) and immature fruit count ( $r = 0.84$ ,  $p < 0.05$ ) was observed. The correlation between visual ratings and the mature ratio was also strong ( $r = 0.84$ ,  $p < 0.05$ ). Conversely, there were strong negative correlations between visual ratings and both immature fruit count ( $r = -0.80$ ,  $p < 0.05$ ) and total fruit count ( $r = -0.61$ ,  $p < 0.05$ ). Using PCA based on these traits, we investigated the contributions of each trait, and the new variable 1, which explains 71% of the variance, is largely influenced by mature, immature, and total fruit count (Figure 6A). Both visual ratings and model-predicted mature ratios are negatively associated with the new variable 2, which explains 23.2% of the variance. We clustered genotypes based on the PCA results into four groups (Figure 6B) and computed the averages of the traits for each group (Table 3). Approximately 28% of the genotypes were grouped into Cluster 1, characterized by a high mature ratio but a low fruit count. These lines, categorized as early maturing, likely had most of their ripened fruits drop to the ground and go undetected, with the image analysis capturing only the remaining ripened fruits on the bushes. Imaging these genotypes earlier could provide a more apparent characterization of their productivity. In contrast, Cluster 3 (24% of the genotypes) had a lower mature ratio (0.63) but a high fruit count, indicating that these genotypes are entering their optimal harvest window and exhibiting strong productivity. Clusters 2 and 4 showed medium fruit counts but differed in mature ratio. Genotypes in Cluster 4 are likely late maturing, with a lower mature ratio (0.55), further

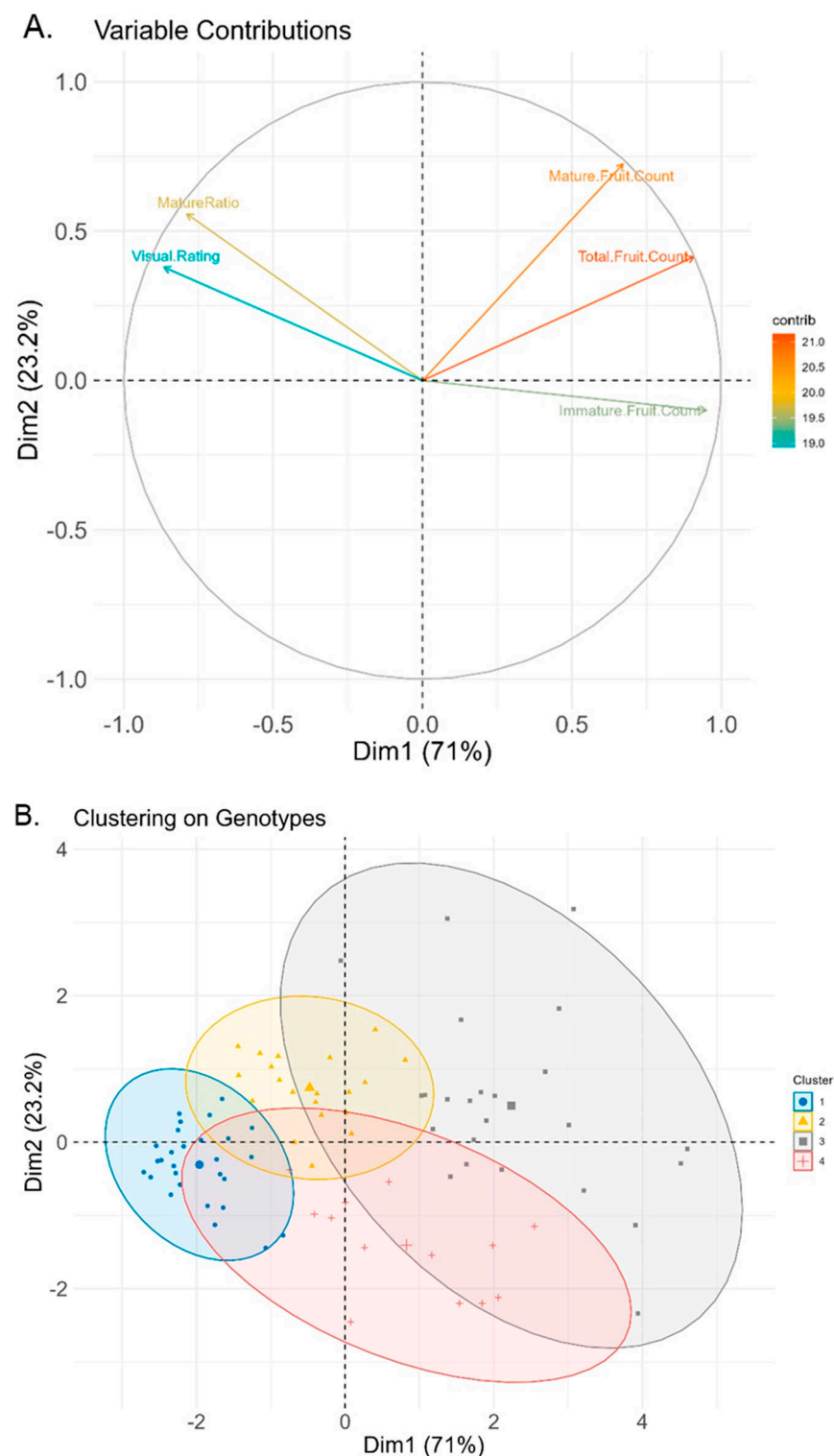
emphasizing the variations in maturity timing across clusters. Since the images were taken at a single time point, direct comparisons of total yield across multiple lines are not feasible. Ongoing work focuses on capturing fruit imaging at multiple time points throughout the season to accurately assess fruit yield. In addition, detecting fruit yield across multiple seasons will provide insights into the environmental impacts on the blueberry yield and time to maturity.



**Figure 5.** Pearson correlation among immature fruit count, mature fruit count, total fruit count, mature ratio detected by the model and visual ratings of mature ratio in a blueberry mapping population in Alapaha, GA, 2023. \*\*, and \*\*\* indicate significance at 0.05, 0.01, and 0.001 *P* level.

**Table 3.** Principal component analysis shows averages of immature, mature, and total fruit count, as well as mature ratio, from model prediction and visual ratings of different clusters.

| Cluster | Genotype% | Fruit Count |        |       | Mature Ratio |               |
|---------|-----------|-------------|--------|-------|--------------|---------------|
|         |           | Immature    | Mature | Total | Model        | Visual Rating |
| 1       | 28        | 21          | 106    | 127   | 0.85         | 5.9           |
| 2       | 20        | 86          | 332    | 418   | 0.80         | 5.6           |
| 3       | 24        | 281         | 457    | 738   | 0.63         | 3.8           |
| 4       | 13        | 160         | 187    | 347   | 0.55         | 3.5           |



**Figure 6.** Principal component analysis (PCA) showing (A) variable contributions for berry count, mature ratio, and visual ratings of maturity and (B) cluster results based on the PCA analysis.

### 3.4. Limitations of the Model and Future Direction for Improvement

The model has some limitations that should be taken into account when interpreting the results. First, the training images were collected exclusively from southern highbush blueberry plants, approximately 4 to 5 years old, and taken in May. This may limit the model's performance when applied to different blueberry types, such as rabbiteye

or northern highbush varieties, particularly at various growth stages. Additionally, the model's detection capabilities only capture a fraction of the total yield (~25%), making it useful in production systems where canopy architectures and growth stages are uniform. While images taken in direct sunlight were included in the training set, the model struggled to identify berries affected by sun glare. Using a polarized filter may help reduce this glare and improve detection accuracy. A few image collection considerations could be implemented to optimize the model's effectiveness, such as capturing images on overcast days, using polarized lenses to minimize glare, and maintaining a consistent distance from the bushes.

Nevertheless, this open access model offers blueberry breeders an image analysis tool that can classify and count immature and mature blueberries in a high-throughput and objective manner. For breeding programs hosting hundreds or even thousands of breeding lines, implementing this new model will replace conventional categorical visual ratings of yield and berry developmental stages by actual berry counts. This valuable information on productivity and maturity will improve the precision of selecting high-yielding and early maturing varieties. In addition, the growers will be able to properly gauge the harvest windows of their fields using this image analysis tool. Future improvements will focus on expanding the model's adaptability to diverse blueberry types and environmental conditions, enhancing its utility across a broader range of applications. In addition, developing an automated image acquisition system would streamline the entire phenotyping workflow, significantly improving the efficiency of data collection.

#### 4. Conclusions

In this study, we developed and shared a high-throughput phenotyping method using neural networks to predict blueberry time to maturity and to provide a proxy for yield. By publishing both the labeled image dataset and the trained model, we aim to facilitate further research in computer vision and precision agriculture. The dataset provided can be easily integrated into other machine learning models, broadening its utility across related fields. Moreover, our model offers practical applications for blueberry breeders, enabling more efficient selection and for growers, helping them accurately estimate optimal harvest windows. This open-source resource can potentially enhance research capabilities and agricultural productivity. However, the generalizability of the model is limited by the diversity of the training dataset, which may not fully capture the variability in different regions or conditions. Efforts should prioritize dataset expansion and diversity, ensuring broader representation of environments and cultivars to improve robustness. Collaboration with breeders and growers is also essential to ensure the model addresses practical challenges and integrates seamlessly into existing agricultural workflows. Additionally, optimizing the model for edge computing and deployment can enable real-time, on-farm decision making.

**Supplementary Materials:** The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/horticulturae10121332/s1>, Figure S1: Metrics during the training process; Figure S2: The relationship between precision, recall, and confidence threshold.

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